ANITA MADANI

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SKILLS

Coding

Python (50k+ lines) | C (10k) | C++ (10k) | Bash (5k) | Java (5k) | MATLAB (3k) | Verilog (1k)

Expertise

Computer Vision | 3D Reconstruction | Visual Localization | Change Detection | Remote Sensing | Deep Learning | Diffusion/Generative Models | LLMs | Multi-Modal Learning | Algorithms & DS | Optimization | Linear Algebra | Probability | Signal & Image Processing

Tools

PyTorch | TensorFlow | Keras | scikit-learn | Hugging Face | ONNX | OpenCV | COLMAP | torchvision | TensorBoard | MLflow | Git | Docker | SLURM | Conda/Mamba | CMake | Linux | AWS | SQL

PROFESSIONAL EXPERIENCE

Machine Learning Research Intern (MLR)

Apple – Video Computer Vision (VCV) Team

May 2026 – Sep 2026 SunnyVale, CA

- Conducting research on video understanding and vision language models within the Video Computer Vision (VCV) team.
- Designing scalable deep learning architectures for multimodal representation learning and temporal reasoning.

Research intern

NEC Laboratories America INC - Media Analytics Lab

Feb 2026 - May 2026

- Architecting generative agent-based AI models using diffusion frameworks to create high-fidelity, reactive simulations for realistic scene synthesis and trajectory forecasting.
- Developing data-driven frameworks for modeling pedestrian dynamics and large-scale human behavior patterns to improve autonomous vehicle predictability.

Graduate Research Assistant

Johns Hopkins University, AIEM & VIU

2024 – Present Baltimore, MD

- Developed deep learning models for **image registration**, **visual localization**, and **3D reconstruction** under challenging real-world conditions.
- Researched **remote sensing** and **change detection** using multimodal representation learning to resolve temporal and seasonal variations.
- Built **diffusion** and **vision-language models** for large-scale scene understanding; collaborated on the multi-institutional WRIVA project.

HONORS & AWARDS

- **Graduate Fellowship** — Whiting School of Engineering, Johns Hopkins University
- **Qualified for national Math and Informatics Olympiad** (2016)
- **Valedictorian** — Farzanegan 1 School (NODET, Middle & High School)

EDUCATION

Ph.D., Electrical & Computer Engineering

Johns Hopkins University

2023 – 05/2028 (expected) United States

MSE, Electrical & Computer Engineering

Johns Hopkins University

2023 - 2025 United States

GPA: 3.92

B.Sc., Electrical Engineering

Sharif University of Technology

2019 – 2023 Iran

GPA: 3.74

PUBLICATIONS & SUBMISSIONS

- **Madani, S.** et al. *DiffRegCD: Integrated Scene Alignment and Change Detection Using Diffusion Features*. WACV 2026.
- **Madani, S.** et al. *Morphing Through Time: Diffusion-Based Bridging of Temporal Gaps for Robust Alignment in Change Detection*. WACV 2026.
- **Gerami, A.** et al. *On The Application of Linear Attention in Multimodal Transformers*.CVPRW 2026.
- **Madani, S.** et al. *PedEditor: Pedestrian Trajectory and Behavior Editing via Generative Models Under Review*,ICLR 2027.
- **Gerami, A.** et al. *IceBerg Attention: Sublinear Long-Context Decoding With Mixture-Of-Gaussians Under Review*, ICLR 2027.

SELECTED PROJECTS & RESEARCH

WRIVA

Johns Hopkins University / IARPA

2024 – Present

- Research on **image registration**, **visual localization**, and contributions to **3D reconstruction** for large-scale, real-world datasets.
- Developed algorithms robust to viewpoint shifts, doppelgänger scenes, and diverse camera models using multi-modal data (ground, UAV, satellite).

Search Engine Optimization via Semantic Graphs

2021 – 2022

- Built a Persian semantic graph with **2M+ documents**, applying GNNs (GCN, GAT) for improved text representations.
- Boosted retrieval with spectral clustering and PageRank, cutting query latency by **23%** and raising MRR by **17%**.

ACADEMIC SERVICE

- **Reviewer**: International Journal of Computer Vision (**IJCV**), IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**) Workshops